

✓ SLIP – 6

Section I: Data Structures [15 Marks]

Q 1) Sort a random array of n integers (accept the value of n from user) in ascending order by using selection sort algorithm.

Answer:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int a[100], n, i, j, min, temp;

    printf("Enter n: ");
    scanf("%d", &n);

    if(n <= 0 || n > 100)
    {
        printf("n must be between 1 and 100");
        return 0;
    }

    for(i = 0; i < n; i++)
        a[i] = rand() % 100;

    printf("Original array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    for(i = 0; i < n - 1; i++)
    {
        min = i;
        for(j = i + 1; j < n; j++)
        {
            if(a[j] < a[min])
                min = j;
        }
        temp = a[i];
        a[i] = a[min];
        a[min] = temp;
    }

    printf("\nSorted array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);
}
```

```
    return 0;
}
```

Section II: Database Management Systems [15 Marks]

Q 2) Consider the following database(primary keys are underlined).

Student (rno,name,city)

Teacher(tno,tname,phone_no,salary)

The relationship is as follows: Student-Teacher: many-to-many with subject as a descriptive attribute.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```
CREATE TABLE Student(
rno INT PRIMARY KEY,
name VARCHAR(50),      city
VARCHAR(50)
);

CREATE TABLE Teacher(
tno INT PRIMARY KEY,
tname VARCHAR(50),
phone_no VARCHAR(15),
salary NUMERIC(10,2)
);

CREATE TABLE Student_Teacher(
    rno INT REFERENCES Student(rno),
    tno INT REFERENCES Teacher(tno),
    subject VARCHAR(50),      PRIMARY
KEY(rno, tno, subject)
);

INSERT INTO Student VALUES
(1,'Amit','Ahilyanagar'),
(2,'Neha','Sangamner'), (3,'Raj','Pune');

INSERT INTO Teacher VALUES
(1,'Mr. Patil','9822000001',45000),
(2,'Prof. Patil','9822000002',55000),
```

```
(3,'Dr. Mulay','9834000003',65000),  
(4,'Dr.Wani','9834000004',35000);
```

```
INSERT INTO Student_Teacher VALUES  
(1,1,'DBMS'),  
(2,2,'IKS'),  
(3,3,'Electronics'),  
(1,4,'IKS'),  
(2,1,'C');
```

Q 2(B)

B. Execute the following queries in PostGreSQL (any 3) [6 marks]

i. List the names of students from 'Ahilyanagar' city.

Answer:

```
SELECT name FROM Student  
WHERE city = 'Ahilyanagar';
```

ii. Display the count of teachers whose salary is between 40000 to 60000.

Answer:

```
SELECT COUNT(*) FROM Teacher  
WHERE salary BETWEEN 40000 AND 60000;
```

iii. Change the phone number of 'Prof. Patil' to '9822131226'

Answer:

```
UPDATE Teacher  
SET phone_no = '9822131226'  
WHERE tname = 'Prof. Patil';
```

iv. List the details of the teachers in the sorted order of their name.

Q 2(C)

C. Create a view to list the details of students who have taken 'DBMS' subject.

Answer:

```
CREATE VIEW Students_Taken_DBMS AS  
SELECT DISTINCT S.*  
FROM Student S
```

```
JOIN Student_Teacher ST ON S.rno = ST.rno  
WHERE ST.subject = 'DBMS';
```

Q 3) Viva [5 marks